

PullUp - Software Requirements Specification (Requirement_v1)

Project Overview

PullUp is a campus-exclusive ride sharing and ride pooling platform designed for Atlas SkillTech University students. The platform aims to reduce transportation costs, improve convenience, increase safety, and enable trusted ride sharing among verified students.

Business Objectives

Reduce commuting costs, improve ride utilization, provide a trusted ecosystem, support ride pooling, and create a scalable mobility platform.

Stakeholders

Students, Founder, Technical Team, External Developers, University Community, Future Campus Partners.

Functional Requirements

FR-001 Student Authentication
FR-002 Profile Management
FR-003 Ride Creation
FR-004 Ride Discovery
FR-005 Ride Pool Formation
FR-006 In-App Chat
FR-007 Notifications
FR-008 Destination Clustering
FR-009 Ride Matching Engine
FR-010 Admin Monitoring

Taxi Pool Module

Students travelling to similar destinations can join shared rides, coordinate via chat, and split transportation costs.

Ride Matching Logic

The preferred approach is destination clustering rather than exact destination matching to increase ride formation success rates.

Technology Stack

Frontend: React Native. Backend: Firebase. Services: Firebase Authentication, Firestore, Cloud Functions, Firebase Cloud Messaging. Maps Integration through Google Maps.

Non-Functional Requirements

NFR-001 Security
NFR-002 Scalability
NFR-003 Availability
NFR-004 Performance
NFR-005 Maintainability

NFR-006 Usability
NFR-007 Reliability

Risks

Developer dependency, scalability concerns, user adoption challenges, ride matching quality, and operational growth challenges.

Roadmap

Phase 1 Atlas Launch, Phase 2 Taxi Pool Optimization, Phase 3 Monetization, Phase 4 Multi-Campus Expansion, Phase 5 Ride Intelligence & Analytics.

Acceptance Criteria

Verified users can create rides, discover pools, receive notifications, coordinate travel, and participate in trusted ride sharing.

Detailed Requirements Expansion

This document serves as the baseline Requirement_v1 and will be referenced by future change requests, approval records, meeting discussions, and Requirement_v2 revisions. The purpose is to act as the primary project context document for enterprise knowledge management, version tracking, semantic retrieval, and RAG-based document intelligence systems.